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Self-supervised multi-modal imitation learning under skewed trajectory demonstrations^{*}Yawen Zhao^{id}, Yue Chen, Fei Zhu^{id}^{*}

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ABSTRACT

Agent behavior consists of two modalities, state-action trajectories and paired videos. Nevertheless, the absence of viable reward functions in practical workflows often undermines reliable assessment and robust control, a challenge that imitation learning addresses by learning policies directly from expert demonstrations. However, imitation learning heavily relies on expert demonstrations, which are typically characterized by skewed trajectory distributions, leading to mode collapse and insufficient coverage of infrequent behaviors. To address the challenges, we propose a two-stage framework for multi-modal policy acquisition under trajectory distribution skew. Specifically, the first stage performs self-supervised pre-training for future state prediction to capture dynamic features more effectively, providing a strong foundation for subsequent policy learning. In the second stage, we encourage tight clustering of samples from similar modes in the latent space while separating those from different modes with a mode-aware contrastive objective, thus improving multi-modal disentanglement. We further introduce a frequency-aware dynamic scaling factor that reweights mode-conditioned learning signals according to mini-batch mode frequencies, thereby increasing the contribution of underrepresented modes. Beyond state-based metrics, we additionally report a complementary post-hoc assessment using Video-Mode Prototype Consistency Accuracy (VMPCA), which measures whether a generated rollout video is assigned to the expert-video prototype corresponding to its predefined behavior mode. The experimental results demonstrate that our method captured both common and rare expert behaviors from skewed trajectory distributions across diverse multi-modal scenarios. It achieves competitive or superior VMPCA performance, providing complementary evidence that generated rollout videos are frequently assigned to expert-mode prototypes corresponding to their conditioning modes.

1. Introduction

In deployed embodied systems, behavior is routinely recorded and reviewed through two complementary sources, trajectories composed of state-action pairs and the corresponding videos. The comprehension and evaluation of these videos form the practical basis for acceptance testing and auditing, yet many real workflows lack viable reward functions, which limits reliable assessment and robust control (Tang et al., 2025). Reinforcement learning has achieved impressive results on complex tasks (Huang et al., 2024), but its dependence on reward design hinders adoption in settings where reward specification is costly, unsafe, or infeasible. Imitation learning offers an alternative by learning policies directly from expert demonstrations and thereby bypasses the reward bottleneck (Foster et al., 2024).

Since its inception, Generative Adversarial Imitation Learning (GAIL) (Ho & Ermon, 2016) has been widely used in different fields.

Nevertheless, real-world demonstrations such as in robotic control and human locomotion (Yao et al., 2024) usually consist of multiple behavior modes coupled together. This behavioral diversity is not merely a theoretical curiosity; it is central to the adaptability of intelligent agents in real applications. Recently, methods like InfoGAIL (Li et al., 2017) and ACGAIL (Lin & Zhang, 2018) attempt to disentangle different behavior modes by introducing latent variables or an auxiliary classifier, which are sensitive to trajectory distribution skew and require balanced mode coverage to prevent mode collapse (Wang et al., 2024). TripleGAIL (Fei et al., 2020) further designs an auxiliary skill selector to select appropriate behavior modes in different environments by making the skill selector and generator work together. However, some demonstrations rarely occur in reality or are difficult to access, resulting in trajectory distribution skew (Yang et al., 2024). Additionally, the essential distinctions between expert behavior modes are often reflected not in static poses at individual time steps, but in the dynamic evolution of

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features such as joint angles, forces, and velocities over time, which further exacerbates the modeling challenge (Cai et al., 2025). To achieve high-fidelity mimicry, it is necessary to be able to accurately capture and reproduce the changing modes of temporal dynamics. However, most existing methods (Fu et al., 2023) still rely on static state-action snapshots, which makes them difficult to effectively model and reconstruct temporal dynamic processes.

To address these challenges, we propose Self-Supervised Multi-modal Generative Adversarial Imitation Learning (SSM-GAIL), a novel imitation learning framework designed to tackle the problems of temporal modeling and trajectory distribution skew in multi-modal expert demonstrations, by introducing a self-supervised pre-trained stage that has three key innovations. First, prior to the adversarial imitation learning (AIL) stage, we augment the discriminator with a decoder and perform self-supervised pre-training by reconstructing multi-step future states with MSE loss. This pre-training step injects rich temporal information into the learned representations, enabling the model to capture fine-grained dynamic cues that distinguish different behavioral modes and providing a solid foundation for subsequent disentanglement tasks. Second, during AIL, we apply InfoNCE contrastive loss (Koromilas et al., 2024) to the same intermediate features of the discriminator, explicitly promoting intra-modal clustering and inter-modal separation in the latent space. This directly addresses the typical problem of mode confusion and mode collapse, leading to more robust and interpretable multi-modal policy learning, even under distribution skew. Third, to further mitigate the adverse effects of demonstration imbalance, we introduce a frequency-aware Dynamic Scaling Factor (DSF) that reweights mode-conditioned learning signals according to mini-batch mode frequencies. Since the scaling factor is inversely related to the empirical mode count, underrepresented modes receive larger weights during mode-conditioned reward computation, improving the coverage of rare behaviors under skewed demonstrations. We evaluate SSM-GAIL on a suite of challenging multi-modal environments, including Mujoco (Todorov et al., 2012) and PandaGym (Gallouédec et al., 2021). While evaluations in imitation learning often rely on privileged states or rewards, real-world auditing is ultimately video-centric. We therefore additionally report a complementary post-hoc assessment that exports short rollouts as videos and computes Video-Mode Prototype Consistency Accuracy (VMPCA) by assigning generated videos to their nearest expert-mode prototypes in a pretrained video feature space. The predefined mode labels are used only to associate generated rollouts with the corresponding expert-video prototype sets.

The main contributions of this work are as follows:

- We propose a self-supervised pre-training framework that mitigates trajectory distribution skew in multi-modal imitation learning by first learning dynamics-aware representations from expert demonstrations.
- We leverage dynamic behavior features extracted from unlabeled trajectories to enrich both policy and discriminator representations, enabling more comprehensive exploration and coverage of diverse behavior modes.
- We enhance the separability of latent variables and more cleanly disentangle multiple skewed behavior modes by embedding contrastive representation learning into the discriminator, yielding more mode-aware decision boundaries.

The remainder of this paper is organized as follows. In Section 2, we review generative adversarial imitation learning, learning from multi-modal demonstrations, and trajectory distribution skew in demonstrations. In Section 3, we introduce the theoretical foundations of imitation learning and its adversarial extensions. In Section 4, we propose the algorithm SSM-GAIL, followed by a theoretical analysis. Section 5 shows and analyzes the experimental results in detail. The conclusion is illustrated in Section 6.

2. Related work

In this section, we take a look at generative adversarial imitation learning, learning from multi-modal demonstrations, and trajectory distribution skew in demonstrations.

2.1. Generative adversarial imitation learning

Inverse Reinforcement Learning (IRL) (Ng & Russell, 2000) is a pivotal framework in imitation learning that seeks to infer a reward function explaining expert demonstrations and then optimizes a policy with reinforcement learning under the inferred reward, but many IRL algorithms invoke an RL solver inside each iteration, which makes them computationally expensive and prone to brittleness on real systems. Adversarial Imitation Learning (AIL) (Xu et al., 2024) reframes imitation as distribution matching without explicitly reconstructing rewards. It sets up a two-player game between a policy (the generator) and a discriminator that tries to tell expert trajectories from agent rollouts. The policy learns to fool the discriminator, driving the agent's occupancy measure toward the expert's. This adversarial recipe avoids solving a full RL subproblem at every step.

One notable AIL method is GAIL, which is an imitation learning method incorporating Generative Adversarial Networks (GANs) (Goodfellow et al., 2014), developed from IRL. The core idea of GAIL is to directly optimize the policy so that the distribution of generated trajectories matches that of expert demonstrations through adversarial training, without explicit reward modeling. It has been shown that GAIL can have a wide range of applications in complex decision-making domains (Luo et al., 2024).

However, GAIL still faces many real-world challenges, such as multi-modal behaviors and skewed trajectory distributions. Addressing these issues remains a key direction for current research on imitation learning.

2.2. Learning from multi-modal demonstrations

When expert demonstrations exhibit multiple modes, standard GAIL suffers from mode collapse due to its single-mode assumption (Allahyani et al., 2023). To address this, multi-modal imitation learning methods have been proposed that assume expert policies follow multiple modes, which is more aligned with real-world applications (Seker et al., 2022). Based on the assumption of multi-modal imitation learning, the modal collapse problem of GAIL can be alleviated (Liu et al., 2024). Recent advances can be categorized as supervised, unsupervised, and semi-supervised imitation learning methods.

Supervised methods leverage additional modal labels or auxiliary classifiers (Lin & Zhang, 2018), achieving effective mode separation but relying heavily on labeled data (Jiang & Nachum, 2020). Triple-GAIL (Fei et al., 2020) designs an auxiliary skill selector to select appropriate behavior modes in different environments by making the skill selector and generator work together. To provide the generator with a good starting point for imitation learning, BAGAIL (Gu & Zhu, 2024) divides the learning process into a pre-training stage followed by an imitation learning stage. Unsupervised methods exploit the mutual information between latent variables and generated behaviors (Li et al., 2017), enabling mode discovery without labels, yet often rely on prior assumptions about the latent space. InfoGAIL variants that enhance latent representation learning, including VAE-GAIL (Wang et al., 2017) for latent inference, and Burn-InfoGAIL (Kuefler & Kochenderfer, 2018) for burn-in demonstration initialization. To combine the strengths of both paradigms and improve performance in complex, real-world scenarios, semi-supervised frameworks such as Ess-InfoGAIL (Fu et al., 2023) have been proposed.

Despite these advances, existing methods still struggle to capture dynamic features and to disentangle modes driven by trajectory differences.

2.3. Trajectory distribution skew in demonstrations

In the field of machine learning and data mining, the problem of data distribution skew is widespread, especially in applications such as medical diagnosis (Xu et al., 2026) and fraud detection (Boyapati & Aygun, 2025).

The research on handling such skew mainly focuses on the data level, the algorithm level, and the improvement of evaluation metrics. The data level mainly mitigates the effect of category imbalance by adjusting the distribution of training data. Among them, oversampling and under-sampling are common methods. The algorithmic level consists mainly of improving the way the classifiers are trained (Zhao et al., 2023) to make them more robust to skewed data. Traditional accuracy tends to fail under skewed data, so researchers have proposed metrics such as F1-score, AUC-ROC, and PR curves (Thabtah et al., 2020) to measure the performance of models on skewed data more comprehensively.

In recent years, methods based on Self-supervised Learning (SSL) (Mildenberger et al., 2025) have also been introduced into the field of skewed data learning to improve the generalization ability of models (Yang & Xu, 2020). In this work, we incorporate self-supervised learning into a generative adversarial network by adding a future state prediction auxiliary task that separates dynamic features from mode features, improves data efficiency, and enables underrepresented trajectory modes to acquire meaningful representations that mitigate mode forgetting under distribution skew. Furthermore, we introduce a frequency-aware dynamic scaling factor to alleviate the effects of skewed trajectory distributions by reweighting mode-conditioned reward signals according to mini-batch mode frequencies.

3. Preliminaries

Imitation learning aims to learn an agent's policy π_θ to approximate an expert policy π_E using expert demonstrations. We model the agent's policy learning task as a Markov Decision Process (MDP) (Park & Levine, 2023) in imitation learning. This MDP is represented as a quintuple $\langle S, \mathcal{A}, \mathcal{P}, R, \gamma \rangle$, where both the state space S and the action space $\mathcal{A}$ are assumed to be finite sets. The transition probability $\mathcal{P}(s_{t+1}|s_t, a_t)$ denotes the dynamics of the environment, where action a_t is determined by the agent's policy $\pi_\theta(a_t|s_t)$. The discount factor $\gamma \in (0, 1)$ is employed to balance the importance of immediate and future rewards, and the extended real number set $\mathbb{R} \cup \infty$ is introduced to cover all possible values. Specifically, in imitation learning, $R(s, a)$ is often derived from expert demonstrations or inferred via IRL, rather than being manually specified as in traditional RL.

GAIL formulates the IRL as an adversarial game between a generator (policy π_θ) and a discriminator D_ψ . The formal GAIL objective is denoted as:

$$V_{\text{GAIL}}(\pi_\theta, D_\psi) = \min_{\pi_\theta} \max_{D_\psi \in (0,1)^{S \times \mathcal{A}}} \mathbb{E}_{\pi_\theta} [\log D_\psi(s, a)] + \mathbb{E}_{\pi_E} [\log(1 - D_\psi(s, a))] - \lambda \mathcal{H}(\pi_\theta), \quad (1)$$

where $\mathcal{H}(\pi_\theta)$ encourages policy entropy for exploration, while λ is the weighting factor for the entropy regularization term.

In the absence of meaningful representations, GAIL often encounters challenges in learning multi-modal behaviors from unstructured demonstrations. To capture diverse multi-modal expert behaviors, InfoGAIL introduces a latent variable c and maximizes the mutual information $I(c; s, a)$ between c and generated trajectories, enabling the policy to represent multiple behavior modes. The objective function for InfoGAIL can be expressed as:

$$V_{\text{InfoGAIL}}(\pi_\theta, D_\psi, Q) = V_{\text{GAIL}}(\pi_\theta, D_\psi) - \lambda_1 \mathcal{L}_1(\pi_\theta, Q), \quad (2)$$

where the regularization term $\mathcal{L}_1$ is defined as:

$$\mathcal{L}_1(\pi_\theta, Q) = \mathbb{E}_{c \sim p(c)} [\log Q(c|s, a)] + \mathcal{H}(c) \leq I(c; s, a), \quad (3)$$

where Q is an auxiliary network for posterior inference, and λ_1 controls the mutual information regularization.

Building on this, Ess-InfoGAIL further addresses skewed demonstrations and richer latent structures by introducing a learnable latent skill variable z and a semi-supervised mutual information term:

$$V_{\text{Ess-InfoGAIL}}(\pi_\theta, D_\psi, Q) = \min_{\pi_\theta, Q} \max_{D_\psi} \mathbb{E}_{\pi_\theta} [\log D_\psi(s, a)] + \mathbb{E}_{\pi_E} [\log(1 - D_\psi(s, a))] - \lambda_1 \mathcal{L}_1(\pi_\theta, Q) - \lambda_2 \mathcal{L}_2(Q) - \lambda_3 \mathcal{H}(\pi_\theta) - \lambda_4 \mathcal{L}_{\text{RIM}}, \quad (4)$$

where $\mathcal{L}_2(Q) = \mathbb{E}_{z \sim p(z)} [\log Q(z|s, a)] + \mathcal{H}(z)$ encourages the extraction of discriminative latent skills. Here, λ_1 controls the mutual-information regularization term $\mathcal{L}_1(\pi_\theta, Q)$, λ_2 weights the Q -based latent skill consistency term $\mathcal{L}_2(Q)$, λ_3 controls the policy entropy regularization $\mathcal{H}(\pi_\theta)$, and λ_4 weights the RIM regularization term $\mathcal{L}_{\text{RIM}}$. $\mathcal{L}_{\text{RIM}}$ regularizes the learned latent distribution towards a prior:

$$\mathcal{L}_{\text{RIM}}(Q, p(c)) = -\frac{1}{N} \sum_i \mathcal{H}(Q(\hat{z}|s_i, a_i)) - D_{KL}(Q(\hat{z})||p(c)) - R(\cdot), \quad (5)$$

where D_{KL} is the KL divergence, and $R(\cdot)$ denotes any additional regularization terms.

4. Methodology

In this section, we introduce the SSM-GAIL framework as shown in Fig. 1, together with the theoretical proof discussion and analyses.

4.1. Pre-training with self-supervised learning

In multi-modal imitation learning with skewed expert demonstrations, different behavior modes often differ not only in static state-action snapshots but also in their temporal evolution. Moreover, rare modes receive weak training signals in pure AIL, leading to mode collapse. To address both issues, we first pre-train the discriminator's feature extractor h_ϕ to predict short-term future states, which (i) enhances sensitivity to small dynamic differences for mode disentanglement, and (ii) provides smoother, lower-variance rewards for the subsequent AIL stage.

In AIL, the discriminator can be decomposed into a feature extractor $h_\phi : (s, a) \mapsto h$, parameterized by ϕ , and a discriminator head $D_\psi : h \mapsto D_\psi(h)$, parameterized by ψ . The mapping of the complete discriminator from (s, a) to the discriminator output can be written as

$$D_\psi(s, a) = D_\psi(h_\phi(s, a)). \quad (6)$$

However, since the discriminator parameters are randomly initialized, the extracted features $h_\phi(s, a)$ do not yet contain sufficient structural information about the environment dynamics or expert demonstrations. As a result, the discriminator has limited ability to distinguish expert demonstrations from agent trajectories, and the reward signal $r_{\psi, \phi}(s, a) = -\log(1 - \sigma(D_\psi(h_\phi(s, a))))$ received by the policy can be noisy and high-variance. When the policy network π_θ is updated using these unstable rewards, the gradient estimates become unreliable, the variance increases, convergence slows, and training may collapse. At the same time, the features extracted by the discriminator may remain insensitive to environmental dynamics and expert behavior, causing subsequent reward computations to be performed in a suboptimal representation space.

Firstly, we randomly sample a pair of (s_t, a_t) from expert demonstrations and take the future state $\{s_{t+1}, \dots, s_{t+H}\}$ of the successive H steps after it. Then, we warm-start the discriminator feature extractor h_ϕ and the discriminator decoder parameterized by ψ_{dec} on expert demonstrations, so that the feature vector $h_\phi(s_t, a_t)$ retains as much information as possible about the future state over the next H steps by minimizing the

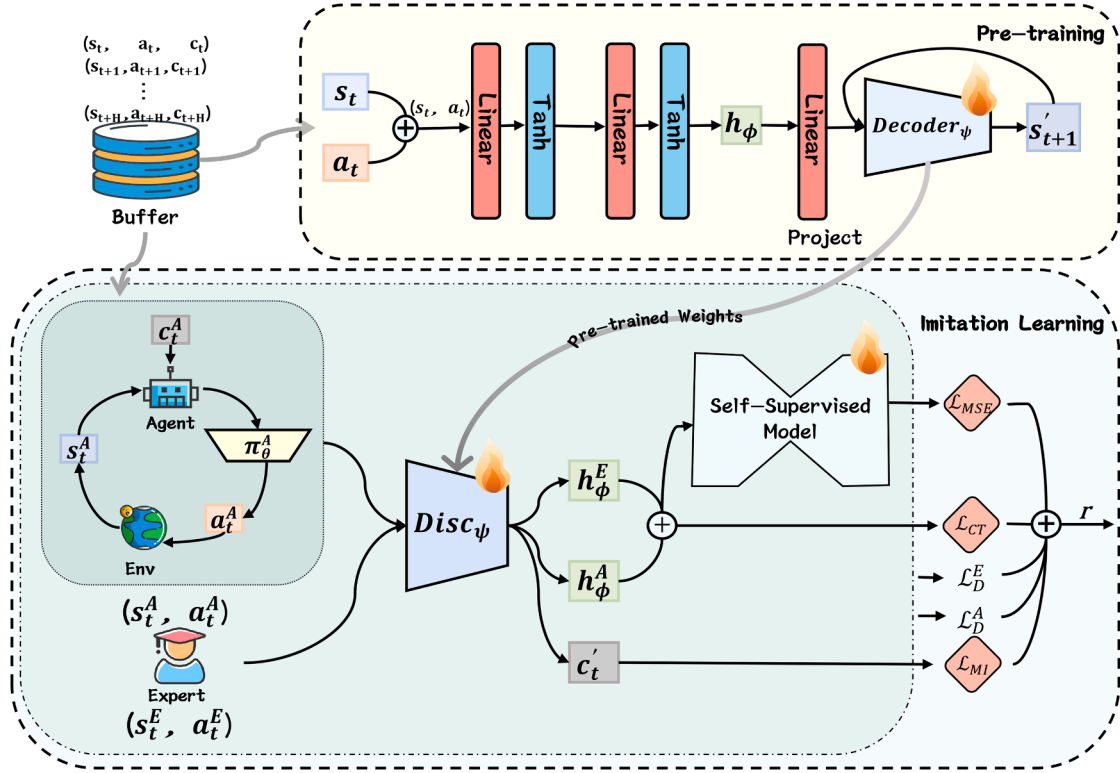


Fig. 1. Overview of the SSM-GAIL Framework. The feature extractor encodes state-action pairs into latent representations, which are first trained in a self-supervised pre-training stage to capture dynamic information over a prediction horizon H . In the imitation learning stage, pre-trained weights are transferred from the self-supervised model to enhance representation learning. The agent interacts with the environment using a policy conditioned on latent behavior modes. Both agent and expert trajectories are processed by the discriminator to extract feature embeddings.

following reconstruction loss:

$$\mathcal{L}_{MSE}(\phi, \psi_{dec}) = \mathbb{E}_{(s_t, a_t) \sim \pi_E} \left\| \text{decoder}_{\psi_{dec}}(h_\phi(s_t, a_t)) - \text{vec}(s_{t+1:t+H}) \right\|^2, \quad (7)$$

where $h_\phi(s_t, a_t) \in \mathbb{R}^d$ is a low-dimensional vector representing features extracted from the current state-action pair containing information that affects the next change in the environment. The prediction horizon H determines the temporal extent of the self-supervised reconstruction target. A smaller horizon emphasizes local transition smoothness and short-term dynamics, whereas a larger horizon encourages the feature extractor to encode longer temporal dependencies. However, increasing H also enlarges the prediction target and increases the uncertainty of future-state reconstruction, which may make the auxiliary task harder to optimize. Since the decoder predicts the future state segment $\text{vec}(s_{t+1:t+H})$ directly from $h_\phi(s_t, a_t)$, H mainly controls the trade-off between temporal informativeness and reconstruction difficulty. By the usual variational bound (Poole et al., 2019), this maximizes the mutual information between the representation and the future state, denoted by:

$$I(h_\phi(s_t, a_t); s_{t+1:t+H}) = H(s_{t+1:t+H}) - H(s_{t+1:t+H} | h_\phi(s_t, a_t)). \quad (8)$$

This objective forces h_ϕ to encode dynamic information. Via the Data-Processing Inequality (Beaudry & Renner, 2012), this also increases $I(c; h)$, sharpening latent code separation and mitigating mode overlap.

We define the maximum error as:

$$\epsilon = \sup_{s_t, a_t} \|h_\phi(s_t, a_t) - h^*(s_t, a_t)\|, \quad (9)$$

where $h^*(s_t, a_t)$ denotes an ideal mapping capable of extracting information related to the state of a future H -steps without loss.

Definition 1 (Lipschitz mapping). A mapping $f : X \rightarrow Y$ is called Lipschitz continuous if there exists a constant $L \geq 0$ such that

$$\|f(x) - f(x')\| \leq L \|x - x'\|, \quad \forall x, x' \in X. \quad (10)$$

Here, L denotes the Lipschitz constant of f , which upper-bounds the rate at which the output of f can change with respect to its input.

As defined in Eq. (9), ϵ is the maximum representation error between h_ϕ and h^* . The following bound is formulated in the learned feature space. Specifically, it analyzes how the representation error ϵ propagates through the discriminator logit head, rather than requiring global Lipschitz continuity of the entire discriminator or feature extractor with respect to the raw state-action input. Let $d(h) = w^\top h + b$ denote the discriminator head. Since this mapping is linear in h , its Lipschitz constant with respect to the feature input can be written as $L_d = \|w\|_2$. During training, the discriminator objective includes a gradient penalty on expert state-action samples, logit-weight regularization, and weight decay, which discourage excessive local gradients and uncontrolled weight growth on the sampled data manifold. Thus, L_d is used as a finite effective head-Lipschitz constant in the subsequent stability analysis.

Lemma 1 (Logit deviation). Let $d_\phi(s_t, a_t) = w^\top h_\phi(s_t, a_t) + b$ and $d^*(s_t, a_t) = w^\top h^*(s_t, a_t) + b$. If the discriminator head has a finite effective Lipschitz constant L_d with respect to its feature input, then the logit deviation induced by the representation error ϵ satisfies

$$|d_\phi(s_t, a_t) - d^*(s_t, a_t)| \leq L_d \epsilon. \quad (11)$$

Since the discriminator head is linear in the feature input, for any two feature vectors u and u' ,

$$|w^\top u + b - (w^\top u' + b)| = |w^\top (u - u')| \leq \|w\|_2 \|u - u'\|_2. \quad (12)$$

Thus, the discriminator head is Lipschitz continuous with respect to the feature input, and its effective Lipschitz constant can be written as

$L_d = \|w\|_2$ under the regularized parameter scale during training. Substituting $u = h_\phi(s_t, a_t)$ and $u' = h^*(s_t, a_t)$ gives

$$\begin{aligned} |d_\phi(s_t, a_t) - d^*(s_t, a_t)| &\leq L_d \|h_\phi(s_t, a_t) - h^*(s_t, a_t)\| \\ &\leq L_d \epsilon. \end{aligned} \quad (13)$$

In the adversarial imitation learning, we normally convert logit to pseudo-probability via sigmoid $\sigma(z) = 1/(1 + e^{-z})$ again, and define the imitation reward r_ϕ and the ideal reward r^* as

$$\begin{aligned} r_\phi(s_t, a_t) &= -\log(1 - \sigma(d_\phi(s_t, a_t))), \\ r^*(s_t, a_t) &= -\log(1 - \sigma(d^*(s_t, a_t))). \end{aligned} \quad (14)$$

As defined in Eq. (14), both r_ϕ and r^* share the form $r(z) = -\log(1 - \sigma(z))$ for different z . Therefore, it suffices to analyze the Lipschitz continuity of $r(z)$ on $\mathbb{R}$, which can be shown to have a finite Lipschitz constant L_r .

Proof. First, the $r(z)$ reduces to the common softplus form:

$$\begin{aligned} 1 - \sigma(z) &= \frac{e^{-z}}{1 + e^{-z}} \\ \Rightarrow r(z) &= -\log \frac{e^{-z}}{1 + e^{-z}} \\ &= \log(e^z + 1). \end{aligned}$$

Thus, we only need to show that $f(z) = \log(1 + e^z)$ is in $\mathbb{R}$ on Lipschitz and find its constant.

f is derivable and its first-order derivative is:

$$f'(z) = \frac{d}{dz} \log(1 + e^z) = \sigma(z).$$

Note that for all z , there is $0 < \sigma(z) < 1$:

$$\forall z \in \mathbb{R} : |f'(z)| \leq 1.$$

For any $z_1, z_2 \in \mathbb{R}$, it follows from the Mean-Value Theorem for real-valued functions that there exists ξ lying in the interval $[z_1, z_2]$ (or $[z_2, z_1]$), we have:

$$\begin{aligned} f(z_1) - f(z_2) &= f'(\xi)(z_1 - z_2) \\ \Rightarrow |f(z_1) - f(z_2)| &= |f'(\xi)| |z_1 - z_2| \\ &\leq 1 \cdot |z_1 - z_2|. \end{aligned}$$

Such that $r(z) = f(z)$, we obtain:

$$|r(z_1) - r(z_2)| \leq |z_1 - z_2|, \quad \forall z_1, z_2 \in \mathbb{R}.$$

Therefore, $r(z)$ is Lipschitz continuous on $\mathbb{R}$, which can be taken as a constant:

$$L_r = \sup_{z \in \mathbb{R}} |r'(z)| = \sup_{z \in \mathbb{R}} \sigma(z) = 1.$$

□

Proving that the function $r(z) = -\log(1 - \sigma(z))$ on $\mathbb{R}$ has a finite Lipschitz constant L_r , we have that:

$$\begin{aligned} |r_\phi(s_t, a_t) - r^*(s_t, a_t)| &= |r(d_\phi(s_t, a_t)) - r(d^*(s_t, a_t))| \\ &\leq L_r |d_\phi(s_t, a_t) - d^*(s_t, a_t)| \\ &\leq L_r L_d \epsilon. \end{aligned} \quad (15)$$

Lemma 2 (Policy gradient error bound). For a fixed policy π_θ , let $Q_\phi^{\pi_\theta}(s_t, a_t)$ denote the state-action value function induced by the discriminator reward r_ϕ . It satisfies the discounted Bellman equation for $\gamma \in [0, 1]$:

$$\begin{aligned} Q_\phi^{\pi_\theta}(s_t, a_t) &= r_\phi(s_t, a_t) \\ &+ \gamma \mathbb{E}_{s_{t+1} \sim P(\cdot | s_t, a_t), a_{t+1} \sim \pi_\theta(\cdot | s_{t+1})} [Q_\phi^{\pi_\theta}(s_{t+1}, a_{t+1})]. \end{aligned} \quad (16)$$

The policy-gradient estimator can be written as

$$\begin{aligned} \nabla_\theta J_\phi(\pi_\theta) &= \mathbb{E}_{(s_t, a_t) \sim \pi_\theta} [\nabla_\theta \log \pi_\theta(a_t | s_t) \cdot Q_\phi^{\pi_\theta}(s_t, a_t)], \\ \|\nabla_\theta \log \pi_\theta(a_t | s_t)\| &\leq C. \end{aligned} \quad (17)$$

where the log-probability gradient is assumed to be bounded by a constant C over the sampled state-action domain during training.

Since the reward error $r_\phi - r^*$ is controlled by $L_r L_d \epsilon$, the induced policy-gradient deviation under the same fixed policy π_θ satisfies

$$\begin{aligned} \|\nabla_\theta J_\phi(\pi_\theta) - \nabla_\theta J^*(\pi_\theta)\| &\leq C \sup_{s_t, a_t} |Q_\phi^{\pi_\theta}(s_t, a_t) - Q^{*, \pi_\theta}(s_t, a_t)| \\ &= O(\epsilon). \end{aligned} \quad (18)$$

Proof. The reward error obtained from Eq. (15) is

$$|r_\phi(s_t, a_t) - r^*(s_t, a_t)| \leq L_r L_d \epsilon, \quad L_r = 1, \quad L_d < \infty.$$

We define the reward error upper bound δ_r and the value-error function $\Delta_Q(s_t, a_t)$ as

$$\begin{aligned} \delta_r &:= \sup_{s_t, a_t} |r_\phi(s_t, a_t) - r^*(s_t, a_t)| \\ &\leq L_r L_d \epsilon = O(\epsilon), \\ \Delta_Q(s_t, a_t) &:= Q_\phi^{\pi_\theta}(s_t, a_t) \\ &\quad - Q^{*, \pi_\theta}(s_t, a_t), \\ \|\Delta_Q\|_\infty &:= \sup_{s_t, a_t} |\Delta_Q(s_t, a_t)|. \end{aligned}$$

Subtracting the Bellman equations associated with r_ϕ and r^* under the same policy π_θ gives

$$\begin{aligned} \Delta_Q(s_t, a_t) &= r_\phi(s_t, a_t) - r^*(s_t, a_t) \\ &\quad + \gamma \mathbb{E}_{s_{t+1} \sim P(\cdot | s_t, a_t), a_{t+1} \sim \pi_\theta(\cdot | s_{t+1})} [\Delta_Q(s_{t+1}, a_{t+1})]. \end{aligned}$$

Taking the supremum over all state-action pairs (s, a) yields

$$\begin{aligned} \|\Delta_Q\|_\infty &\leq \delta_r + \gamma \|\Delta_Q\|_\infty, \\ (1 - \gamma) \|\Delta_Q\|_\infty &\leq \delta_r, \\ \|\Delta_Q\|_\infty &\leq \frac{\delta_r}{1 - \gamma} = O(\epsilon). \end{aligned}$$

Therefore,

$$\sup_{s_t, a_t} |Q_\phi^{\pi_\theta}(s_t, a_t) - Q^{*, \pi_\theta}(s_t, a_t)| = O(\epsilon).$$

According to Eq. (17), we define the policy-gradient deviation as

$$\begin{aligned} \Delta_G &:= \nabla_\theta J_\phi(\pi_\theta) - \nabla_\theta J^*(\pi_\theta) \\ &= \mathbb{E}_{(s_t, a_t) \sim \pi_\theta} [\nabla_\theta \log \pi_\theta(a_t | s_t) \Delta_Q(s_t, a_t)]. \end{aligned}$$

Taking the ℓ_2 norm and using Jensen's inequality gives

$$\begin{aligned} \|\Delta_G\|_2 &\leq \mathbb{E}_{(s_t, a_t) \sim \pi_\theta} \left[\left\| \nabla_\theta \log \pi_\theta(a_t | s_t) \Delta_Q(s_t, a_t) \right\|_2 \right] \\ &\leq \mathbb{E}_{(s_t, a_t) \sim \pi_\theta} \left[\left\| \nabla_\theta \log \pi_\theta(a_t | s_t) \right\|_2 |\Delta_Q(s_t, a_t)| \right] \\ &\leq C \mathbb{E}_{(s_t, a_t) \sim \pi_\theta} [|\Delta_Q(s_t, a_t)|] \\ &\leq C \sup_{s_t, a_t} |\Delta_Q(s_t, a_t)| \\ &= C \|\Delta_Q\|_\infty = O(\epsilon). \end{aligned}$$

□

When the self-supervised loss $\mathcal{L}_{MSE}$ is sufficiently small, the representation error ϵ is also reduced. Consequently, the reward-induced policy-gradient deviation is controlled at the order of $O(\epsilon)$, which helps reduce the variance of policy updates and improves training stability.

In this way, the discriminator can provide clearer and more stable reward signals when entering the adversarial imitation stage, thereby reducing policy-update variance, accelerating convergence, and improving the final policy performance.

4.2. Contrastive learning to improve disentanglement

To enhance the discriminator's intermediate representation h for separating distinct latent variables c , we introduce a mode-aware contrastive regularization term during adversarial training. The loss promotes clustering of features from the same mode and separation of features from different modes, thereby improving AIL's ability to recover multi-modal expert behavior.

We denote $\{(h_i, y_i)\}_{i=1}^B$ as a minibatch of mode-identified features, where $y_i \in \{1, \dots, K\}$ is the mode identifier used to construct the contrastive mask. For policy-generated samples, y_i is induced by the latent mode c_i sampled from the prior $p(c)$, since each rollout is generated by a policy conditioned on c_i . Let $s_{ij} = h_i^\top h_j$ denote the similarity between features h_i and h_j . The logits are computed as $\ell_{ij} = s_{ij}/\tau$. The same-mode mask is defined as

$$m_{ij} = \begin{cases} 1, & y_i = y_j \text{ and } i \neq j, \\ 0, & \text{otherwise.} \end{cases}$$

The mask m_{ij} is used to select positive sample pairs.

Given a temperature $\tau > 0$, we define the contrastive loss as:

$$\mathcal{L}_{CT} = \frac{1}{B} \sum_i \left(-\frac{1}{\sum_j m_{ij}} \sum_{j=1}^B m_{ij} \log Q(j|i) \right), \quad (19)$$

$$Q(j|i) = \frac{\exp(\ell_{ij})}{\sum_{k=1}^B \exp(\ell_{ik})},$$

where B is the total number of samples in a minibatch, $Q(j|i)$ is the probability of matching the i th feature to the j th feature, and $\sum_j m_{ij}$ denotes the number of positive samples associated with sample i . The computation of $\mathcal{L}_{CT}$ does not require additional ground-truth mode annotations for unlabeled expert demonstrations. The mask m_{ij} is constructed from available or induced mode identifiers. Available mode identifiers are provided by the small mode-identified subset of expert demonstrations used for mode-supervised signals. For policy-generated samples, the mode identifier is induced by the latent mode c_i sampled from the prior $p(c)$, since each rollout is generated by a policy conditioned on c_i . Unlabeled expert demonstrations are not assigned additional manual mode labels for constructing supervised positive pairs. They continue to contribute to adversarial distribution matching and self-supervised future-state prediction; when their features are included in the adversarial feature space, they are used as weakly paired feature samples for representation regularization rather than as manually labeled expert examples. Therefore, the contrastive term is used as a mode-aware feature regularizer rather than as a procedure that infers manual labels for all unlabeled expert samples.

This mode-aware contrastive regularization can be viewed as an InfoNCE-style objective that encourages features sharing the same available or induced mode identifier to be closer while separating features with different identifiers. It increases the mutual information between the discriminator feature h and the mode identifier y , which can be written as

$$I(h; y) = H(y) - H(y|h) = \mathbb{E}_{p(h,y)} [\log p(y|h)] + H(y), \quad (20)$$

where $H(y) = -\sum_y p(y) \log p(y)$ is the entropy of the discrete variable y , independent of the model parameters.

Definition 2 (Fano Inequality). With a classifier $\hat{Y}(h)$ predicting label y from feature h , we define the error rate to be $P_e = \Pr[\hat{Y}(h) \neq y]$. Fano inequality is written as:

$$H(y|h) \leq H_b(P_e) + P_e \log(K-1), \quad (21)$$

where $H_b(P_e) = -P_e \log P_e - (1-P_e) \log(1-P_e)$ is the binary entropy.

Proof. From the KL dispersion non-negativity:

$$D_{KL}(p(y|h) \| Q(y|h)) = \mathbb{E}_{p(y|h)} \left[\log \frac{p(y|h)}{Q(y|h)} \right] \geq 0,$$

where $Q(y|h)$ is an arbitrary conditional distribution.

It follows that:

$$\begin{aligned} \mathbb{E}_{p(y|h)} [\log p(y|h)] &\geq \mathbb{E}_{p(y|h)} [\log Q(y|h)] \\ \Rightarrow \int p(h) \mathbb{E}_{p(y|h)} [\log p(y|h)] dh &\geq \\ \int p(h) \mathbb{E}_{p(y|h)} [\log Q(y|h)] dh. \end{aligned}$$

The left part of the inequality above:

$$\begin{aligned} &\int p(h) \mathbb{E}_{p(y|h)} [\log p(y|h)] dh \\ &= \int \int p(h) p(y|h) \log p(y|h) dy dh \\ &= \mathbb{E}_{p(h,y)} [\log p(y|h)]. \end{aligned}$$

Similarly, the right part of the inequality becomes:

$$\begin{aligned} &\int p(h) \mathbb{E}_{p(y|h)} [\log Q(y|h)] dh \\ &= \int p(h) \left(\int p(y|h) \log Q(y|h) dy \right) dh \\ &= \int \int p(h) p(y|h) \log Q(y|h) dy dh \\ &= \mathbb{E}_{p(h,y)} [\log Q(y|h)]. \end{aligned}$$

Therefore, the above inequality becomes:

$$\mathbb{E}_{p(h,y)} [\log p(y|h)] \geq \mathbb{E}_{p(h,y)} [\log Q(y|h)].$$

For compactness, let $\mathbb{E}_p$ denote $\mathbb{E}_{p(h,y)}$. Adding the entropy of the discrete variable y to both sides preserves the inequality:

$$\begin{aligned} \mathbb{E}_p [\log p(y|h)] + H(y) &\geq \mathbb{E}_p [\log Q(y|h)] + H(y), \\ I(h; y) &\geq \mathbb{E}_p [\log Q(y|h)] + H(y). \end{aligned}$$

In a minibatch, we draw $i \in \{1, \dots, N\}$ with equal probability, and then draw a positive sample $j (j \neq i)$ uniformly from the set of indexes with the same labels as i . We take $Q(y|h) = Q(j|i)$ as the variational posterior in the Evidence Lower Bound:

$$\begin{aligned} I(h; y) &\geq \mathbb{E}_{\substack{i \sim [1,B] \\ j : y_j = y_i, j \neq i}} [\log Q(j|i)] + H(y) \\ &= \frac{1}{B} \sum_i \left(\frac{1}{\sum_j m_{ij}} \sum_{j=1}^B m_{ij} \log Q(j|i) \right) + H(y) \\ &= -\mathcal{L}_{CT} + H(y), \end{aligned}$$

where $m_{ij} = [y_i = y_j] (i \neq j)$. Thus, minimising the contrastive loss $\mathcal{L}_{CT}$ is equivalent to maximising the mutual information lower bound $I(h; y)$.

From the Fano Inequality, we can deduce:

$$\begin{aligned} I(h; y) &= H(y) - H(y|h) \\ &\geq H(y) - H_b(P_e) - P_e \log(K-1) \\ \Rightarrow P_e &\leq \frac{H_b(P_e) + P_e \log(K-1)}{\log(K-1)} \\ &\approx \frac{H(y) - I(h; y) + 1}{\log(K-1)}. \end{aligned}$$

Fano Inequality (Han & Verdú, 1994) yields a relationship between $H(y|h)$ and the classification error rate P_e . Maximising mutual information makes $H(y|h)$ smaller, namely P_e smaller, thus reducing the confusion of different modes and improving decoupling. $\square$

Intuitively, minimising $\mathcal{L}_{CT}$ will bring h_i, h_j closer together under the same label (by increasing ℓ_{ij}), and push h_i, h_k further away under different labels (by decreasing ℓ_{ik}).

4.3. Dynamic scaling factor for skewed modes

In skewed multi-modal demonstrations, dominant modes appear much more frequently than rare modes, causing mode-conditioned reward and representation learning signals to be dominated by majority

modes. To mitigate this effect, we introduce a Dynamic Scaling Factor (DSF) that reweights the Q -based mode-consistency reward according to the empirical mode frequency within each generated rollout mini-batch.

Let $B_\pi = \{(s_i, a_i, c_i)\}_{i=1}^B$ denote a mini-batch of policy-generated state-action samples, where c_i is the latent mode sampled from the prior $p(c)$ in Algorithm 1. We denote by y_i the discrete mode identifier induced by c_i . For K behavior modes, the empirical count of mode k in the mini-batch is

$$n_k = \sum_{i=1}^B \mathbb{I}[y_i = k], \quad (22)$$

where $\mathbb{I}[\cdot]$ is the indicator function.

The DSF weight for mode k is defined by normalized inverse-frequency weighting:

$$\alpha_k = \frac{K(n_k + \delta)^{-1}}{\sum_{\ell=1}^K (n_\ell + \delta)^{-1}}, \quad (23)$$

where δ is a small numerical stability constant. In our implementation, $\delta = 10^{-5}$. Each generated sample is assigned the weight α_{y_i} according to its induced mode identifier.

Recall that Q denotes the auxiliary posterior network used to infer the latent mode from a state-action pair. Without DSF, the mode-consistency reward induced by Q is proportional to $\log Q(y_i | s_i, a_i)$. With DSF, this term is reweighted as

$$r_Q^{\text{DSF}}(s_i, a_i) = \lambda_2 \alpha_{y_i} \log Q(y_i | s_i, a_i), \quad (24)$$

where λ_2 weights the Q -based mode-consistency term. During policy optimization, the discriminator-derived imitation reward $r_{\psi, \phi}(s_i, a_i)$ is augmented with the DSF weighted mode-consistency reward:

$$\tilde{r}(s_i, a_i) = r_{\psi, \phi}(s_i, a_i) + r_Q^{\text{DSF}}(s_i, a_i). \quad (25)$$

Thus, DSF does not introduce an additional standalone loss term. Instead, it reweights the existing Q -based mode-consistency reward before the PPO update. The weighting is dynamic because α_k is recomputed for every generated rollout mini-batch. It is directly related to the skew ratio: when a mode appears less frequently, its count n_k becomes smaller and its scaling factor α_k becomes larger. In the idealized two-mode case, the relative scaling between a minority mode and a majority mode is approximately proportional to the ratio between their sample counts. Therefore, DSF compensates for trajectory distribution skew by increasing the contribution of underrepresented modes during mode-conditioned reward computation.

4.4. Algorithm description

With the aforementioned improvements, the total objective function of SSM-GAIL eventually becomes:

$$V_{\text{SSM-GAIL}}(\pi_\theta, D_\psi, Q, \phi, \psi_{\text{dec}}) = V_{\text{Ess-InfoGAIL}}(\pi_\theta, D_\psi, Q) - \lambda_{\text{self}} \mathcal{L}_{\text{MSE}}(\phi, \psi_{\text{dec}}) - \lambda_{\text{CT}} \mathcal{L}_{\text{CT}}, \quad (26)$$

where λ_{self} and λ_{CT} are two weighting coefficients. θ, ϕ, ψ , and ψ_{dec} denote the parameters of the policy π_θ , the feature extractor h_ϕ , the discriminator D_ψ , and the discriminator decoder, respectively. The decoder network is denoted by $\text{decoder}_{\psi_{\text{dec}}}$. Here, $\lambda_1, \lambda_2, \lambda_3$, and λ_4 weight the mutual-information regularization term $\mathcal{L}_1(\pi_\theta, Q)$, the Q -based latent skill consistency term $\mathcal{L}_2(Q)$, the policy entropy regularization term $\mathcal{H}(\pi_\theta)$, and the RIM regularization term $\mathcal{L}_{\text{RIM}}$, respectively.

The outline of the training process of SSM-GAIL from multi-modal skewed demonstrations is shown in Algorithm 1. We initialize the required parameters and define the prior distribution of the modes as $c \sim p(c)$. The DSF introduced in Section 4.3 is not an additional standalone loss term; instead, it reweights the existing Q -based mode-consistency reward and augments the discriminator-derived imitation reward before the PPO update. Steps 3–8 perform self-supervised pre-training, using expert demonstrations, we train the feature extractor ϕ to predict future state sequences and capture environmental dynamics. Steps

Algorithm 1 Self-supervised multi-modal imitation learning from skewed trajectory distributions.

Input: Expert unlabeled and labeled demonstrations $\mathcal{R}_E \sim \pi_E$, total imitation iterations T , warm-up steps M , batch size B , prediction horizon H , contrastive temperature τ , weighting coefficients $\lambda_1 \sim \lambda_4, \lambda_{\text{self}}, \lambda_{\text{CT}}$, the prior distribution of the modes $p(c)$

Parameter: Parameters of the policy, the feature extractor, the discriminator and the discriminator decoder $\theta, \phi, \psi, \psi_{\text{dec}}$

Output: Learned policy π_θ

1: **Stage 1: Pre-training**

2: **for** $m = 1$ **to** M **do**

3: Sample a minibatch $\{(s_i, a_i, s_{i+1:i+H})\}_{i=1}^B \sim \mathcal{R}_E$

4: Compute features $h_i \leftarrow h_\phi(s_i, a_i)$

5: Decode future states $\hat{s}_{i+1:i+H} \leftarrow \text{decoder}_{\psi_{\text{dec}}}(h_i)$

6: Compute self-supervised loss $\mathcal{L}_{\text{MSE}}(\phi, \psi_{\text{dec}})$

7: Update ϕ and ψ_{dec} by descending gradients of $\mathcal{L}_{\text{MSE}}(\phi, \psi_{\text{dec}})$

8: **end for**

9: **Stage 2: Imitation Learning**

10: **for** $t = 1$ **to** T **do**

11: Sample a batch of latent variables $c_i \sim p(c)$

12: Set mode identifiers y_i according to c_i

13: Sample trajectories (s_i, a_i) from policy π_θ conditioned on c_i

14: Sample expert state-action pairs from $\mathcal{R}_E$

15: Update the discriminator parameters ψ with gradients according to $V_{\text{SSM-GAIL}}$

16: Extract intermediate features h_ϕ from the discriminator

17: Update ϕ by descending gradients of the contrastive loss $\mathcal{L}_{\text{CT}}$

18: Compute mini-batch mode counts and DSF weights α_k

19: Compute the DSF-weighted mode-consistency reward $r_Q^{\text{DSF}}(s_i, a_i)$

20: Take a policy step from θ_t to θ_{t+1} using PPO with the augmented reward $\tilde{r}(s_i, a_i)$

21: **end for**

11–20 constitute the imitation learning stage. At each iteration, latent variables c_i are sampled from $p(c)$, mode identifiers y_i are induced by c_i , and agent state-action pairs are generated via the policy π_θ . The discriminator representation is updated with contrastive learning, DSF weights are computed from the generated rollout mini-batch, and the policy is updated using PPO with the augmented reward $\tilde{r}(s_i, a_i)$.

5. Experiments

5.1. Experimental setup

Experimental Environments. We validate our method by experimenting with tasks in a variety of environments, including a simple 2D circle drawing task (Li et al., 2017), a PandaGym robotic arm control task (Galloudec et al., 2021), and four MuJoCo robot control tasks (Todorov et al., 2012).

- **2D-Trajectory.** 2D-Trajectory (Fig. 2(a)) represents the motion of an object on a 2D plane, where expert demonstrations include four trajectory modes generated by varying parameters such as center, radius, and motion direction with slight random noise.
- **PandaGym.** PandaReach (Fig. 2(b)) environment uses the PyBullet physics engine that learns to place its end-effector at a target position. The environment is extended to five distinct modes, with each mode corresponding to a unique target position arranged evenly on a circle around the origin in the X-Y plane. The target position for each mode is set at a fixed distance from the origin, and the agent receives only proprioceptive observations without direct access to the goal location.
- **Mujoco.** Reacher (Fig. 2(c)) is a two-jointed robot arm to reach one of six targets placed uniformly around a circle by moving the fingertip. The robot starts from random joint angles and observes

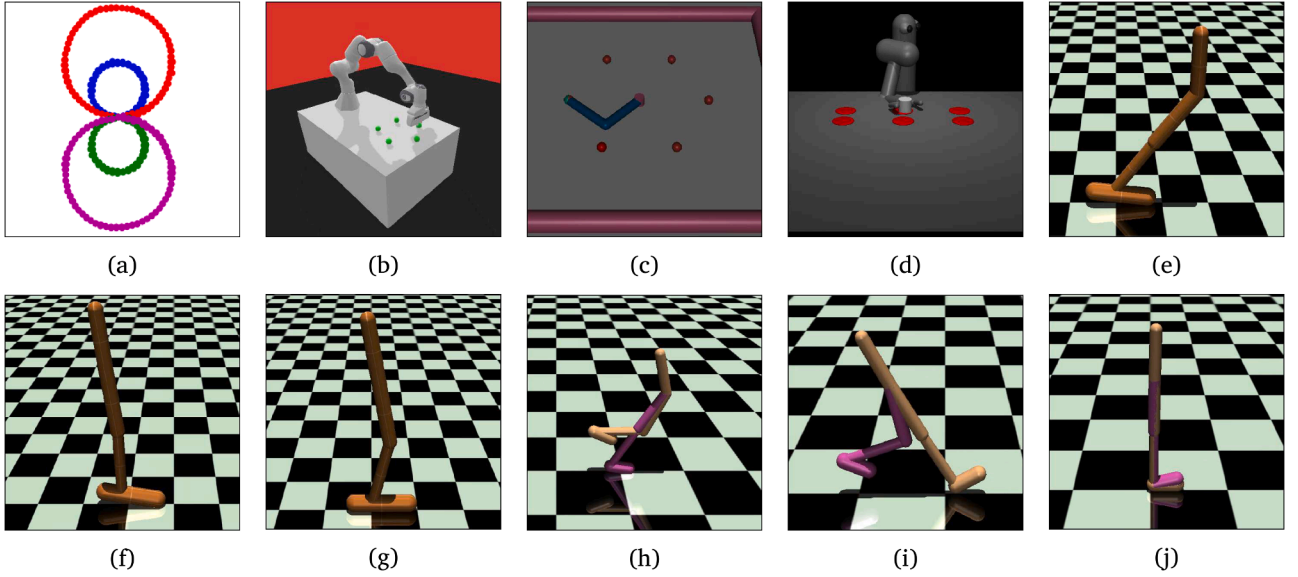


Fig. 2. Image displays of MuJoCo environments and modes. (a) 2D-Trajectory environment, where four colors represent four distinct modes. (b) PandaReach environment, with five green spheres indicating target positions for the end-effector in five different modes. (c) Reacher environment, where six red dots denote six targets, and the robot arm’s endpoint coincides with a red dot to indicate task completion for each mode. (d) Pusher environment, in which six red dots represent six modes, and the robot’s control cylinder must reach each red dot as the target. (e)-(g) Three modes of Hopper: forward, backward, and balancing. (h)-(j) Three modes of Walker2D: forward, backward, and balancing.

only its proprioceptive states, with the target fixed per mode. Pusher (Fig. 2(d)) requires a multi-jointed robot arm to push a cylinder to a goal using the fingertip, relying on proprioception and the positions of its end-effector and the object. Hopper (Fig. 2(e)-(g)) features a one-legged robot with three modes: forward hopping, backward hopping, and balancing, each starting from a randomized state with proprioceptive observations. Walker2D (Fig. 2(h)-(j)) extends Hopper by adding legs, also with three modes: running forwards, running backwards, and balancing.

Expert Demonstration Collection. In our experiments, we trained a separate expert policy for each behavior mode. For data collection, we used each expert policy to interact with the environment and generate expert trajectories. A small portion of the demonstrations is associated with explicit mode identifiers for mode-supervised training signals, while the larger training buffer is used to construct skewed mode distributions for imitation learning. The process of collecting expert demonstrations is illustrated in Fig. 3. For each environment, expert demonstrations are first collected separately from the expert policy of each behavior mode and stored as mode-specific buffers of state-action transition samples. These stored buffers are initially balanced across modes. The skewed demonstration set used for imitation training is then constructed by retaining different numbers of samples from different modes. We define the imbalance ratio ρ as

$$\rho = \frac{\max_k N_k}{\min_k N_k}, \quad (27)$$

where N_k denotes the effective number of transition samples retained for mode k in the skewed training set. Given a stored buffer size N per mode, modes are divided into three frequency groups with effective sample counts

$$N_k = \left\lfloor \frac{N}{\rho^{g/2}} \right\rfloor, \quad g \in \{0, 1, 2\}. \quad (28)$$

Thus, the three groups contain approximately N , $N/\sqrt{\rho}$, and N/ρ samples, respectively. When the number of modes is not divisible by three, the remaining modes are assigned to the higher-frequency groups first. In the main experiments, $\rho = 20$. The stored buffer size is 100,000 samples per mode for Reacher, Pusher, Walker2D, Hopper, and PandaReach,

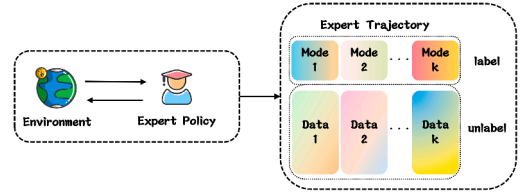


Fig. 3. Schematic of constructing multi-modal expert demonstrations with skewed trajectory distribution (k behavior modes).

and 20,000 samples per mode for 2D-Trajectory. The effective sample counts for each mode are obtained directly from the above rule according to the number of modes in each task. Detailed transition-sample counts, mode-identified-to-large-buffer ratios, effective skewed training-set sizes, and network architecture parameters are provided in Appendix A for reproducibility.

Baselines. We used four baseline methods to demonstrate the efficiency of our method SSM-GAIL. GAIL (Ho & Ermon, 2016) serves as the benchmark for generative adversarial imitation learning. InfoGAIL (Li et al., 2017) extends GAIL by introducing latent variables to distinguish different behavior modes. ACGAIL (Lin & Zhang, 2018) incorporates auxiliary classifiers to improve mode separation. Ess-InfoGAIL (Fu et al., 2023) employs semi-supervised learning to enhance the disentanglement of distinct behavior modes.

Evaluation Metrics. We evaluate the learned policies from complementary perspectives. Normalized Mutual Information (NMI) (Darin et al., 2024) and Average Entropy (ENT) (Lu et al., 2024) are used to assess representation-level behavior mode separation, while Video-Mode Prototype Consistency Accuracy (VMPCA) is reported as a complementary post-hoc video-level metric that measures whether generated rollout videos are assigned to expert-video prototypes corresponding to their predefined behavior modes. Each mode is represented by a discrete mode identifier associated with a specific target, goal, or locomotion behavior. The exact behavior-mode specifications corresponding to the discrete mode identifiers used in the four video-evaluated environments are summarized in Table 1.

Table 1

Exact behavior-mode specifications corresponding to the mode identifiers used in the VMPCA evaluation.

Environment	Mode ID	Mode specification
Reacher	0	Target (0.1500, 0.0000)
	1	Target (0.0750, 0.1299)
	2	Target (−0.0750, 0.1299)
	3	Target (−0.1500, 0.0000)
	4	Target (−0.0750, −0.1299)
Pusher	5	Target (0.0750, −0.1299)
	0	Goal (0.00, 0.00)
	1	Goal (0.00, −0.43)
	2	Goal (0.00, −0.86)
	3	Goal (0.20, 0.00)
Walker2D	4	Goal (0.20, −0.43)
	5	Goal (0.20, −0.86)
	0	Forward locomotion
PandaReach	1	Backward locomotion
	2	Balancing in place
PandaReach	0	Target (0.1500, 0.0000, 0.0250)
	1	Target (0.0464, 0.1427, 0.0250)
	2	Target (−0.1214, 0.0882, 0.0250)
	3	Target (−0.1214, −0.0882, 0.0250)
	4	Target (0.0464, −0.1427, 0.0250)

- **NMI.** NMI quantifies the correspondence between the predicted assignments and the ground-truth labels, normalized to the range [0, 1]. A higher NMI indicates better alignment between the predicted and true modes.
- **ENT.** ENT measures the entropy of the predicted probability distribution for each generated state-action pair. It reflects the purity of behavior modes, where lower ENT indicates purer and more distinct modes.
- **VMPCA.** Video-Mode Prototype Consistency Accuracy (VMPCA) evaluates whether a generated rollout video is assigned to the expert-video prototype associated with the behavior mode used to condition the rollout. The metric operates on pretrained video features and predefined discrete mode labels. Unlike NMI and ENT, which are computed from state-action representations, VMPCA is computed in video feature space. For each rollout video, we uniformly sample 16 frames and extract a 512-dimensional feature using a pretrained R3D-18 video backbone with the classification head removed. The extracted feature is then ℓ_2 -normalized. Let $f(v)$ denote the normalized feature of video v . For each behavior mode k , we compute an expert-mode prototype by averaging the features of the expert videos associated with that mode:

$$\mu_k = \frac{1}{|\mathcal{E}_k|} \sum_{v_i \in \mathcal{E}_k} f(v_i), \quad (29)$$

where $\mathcal{E}_k$ denotes the set of expert videos associated with behavior mode k . For a generated video v_j produced under mode label y_j , we assign it to the nearest expert-mode prototype in Euclidean distance:

$$\hat{y}_j = \arg \min_k \|f(v_j) - \mu_k\|_2. \quad (30)$$

VMPCA is then defined as the fraction of generated videos whose nearest-prototype assignment matches the mode used to condition the rollout:

$$\text{VMPCA} = \frac{1}{N} \sum_{j=1}^N \mathbb{I}[\hat{y}_j = y_j]. \quad (31)$$

A higher VMPCA indicates that generated rollout videos are more frequently assigned to expert-mode prototypes corresponding to their conditioning modes. Therefore, VMPCA complements NMI and ENT by providing a post-hoc video-level assessment of prototype-based behavior-mode consistency.

Hyperparameter. The main hyperparameters used in our experiments are listed in Table 3. These include loss coefficients, learning rates for different components (policy, value, discriminator, encoder, and prior), as well as key algorithmic parameters such as the discount factor γ , prediction horizon H , and coefficients for self-supervised and contrastive learning objectives. Unless otherwise specified, all experiments are conducted using these default values. For all comparative and ablation studies, except those explicitly designed to assess the impact of imbalance, the ratio is fixed at 20.

5.2. Results and analysis

In this section, we present and analyze the experimental results to demonstrate the effectiveness of our proposed method.

5.2.1. Performance comparison

Table 2 presents a comprehensive comparison of all methods in terms of NMI and ENT across six benchmark environments. In tasks characterized by rich multi-modal structure, conventional algorithms such as GAIL and InfoGAIL consistently demonstrate limited disentanglement capacity, frequently failing to capture the full diversity of demonstrated behaviors. Across all environments except Walker2D, SSM-GAIL achieved the highest NMI and lowest ENT. Following cross-task aggregation, SSM-GAIL recorded an average NMI of 0.863 and an average ENT of 0.166. Ess-InfoGAIL followed with an average NMI of 0.830 and average ENT of 0.221; ACGAIL achieved an average NMI of 0.652 and average ENT of 0.465; InfoGAIL recorded an average NMI of 0.609 and average ENT of 0.491; GAIL attained an average NMI of 0.499 and average ENT of 0.544. Compared to the strongest baseline, Ess-InfoGAIL, SSM-GAIL achieved a 3.9% improvement in NMI and a 24.9% reduction in ENT. Relative to the traditional baseline GAIL, NMI improved by 73.0% and ENT decreased by 69.5%. These results demonstrate that incorporating temporal state-space modeling significantly enhances the separability of multi-modal behaviors and yields more definitive mode assignments. Compared to Ess-InfoGAIL, SSM-GAIL demonstrates the most significant improvement in the Reacher environment, with an increase of 0.096 in the NMI metric and a decrease of 0.171 in the ENT metric, effectively mitigating mode confusion in multi-modal scenarios. However, Walker2D stands as the exception, where Ess-InfoGAIL outperforms SSM-GAIL in both NMI and ENT. This phenomenon relates to the strongly periodic contact and long-term dependencies in Walker2D, where reliance solely on temporal latent variables may still prove restrictive. We further examine this limitation specific to Walker2D through the ablation analysis in Section 5.2.5.

Calculating the standard deviation across different random seeds reveals that, across six environments, SSM-GAIL exhibits the lowest standard deviation for NMI (0.056) and ENT (0.054) among all methods. This indicates its greater insensitivity to random initialisation and sampling noise, alongside more stable convergence. It should be noted that SSM-GAIL exhibits a relatively large NMI variance (± 0.174) on Walker2D, consistent with its sensitivity to contact events and phase transitions. Across all data points, NMI generally increases as ENT decreases, indicating that higher clustering quality corresponds to lower code assignment entropy. This validates the consistency of both metrics in evaluating multi-modal retention capability. SSM-GAIL achieves stronger multi-modal retention and lower uncertainty across the vast majority of tasks, while demonstrating superior stability across seeds. The exception of Walker2D indicates that integrating explicit dynamics, phase priors, and temporal latent variables represents a promising direction for further enhancing performance in complex contact dynamics scenarios.

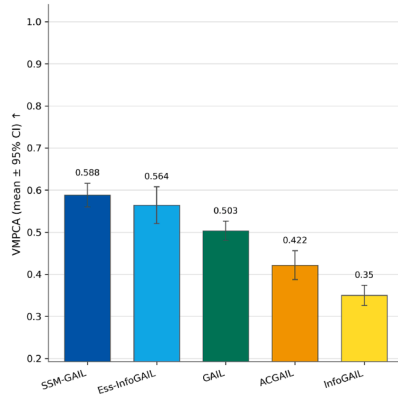
5.2.2. Prototype-based video-mode consistency

Following the VMPCA protocol defined in Section 5.1, Fig. 4 reports prototype-based video-mode consistency across four environments. VMPCA measures whether a generated rollout video is assigned to the

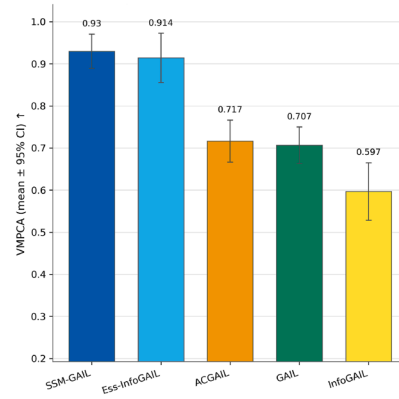
Table 2

Comparison of NMI ($\uparrow$) and ENT ($\downarrow$) for all methods across six benchmark tasks. We report the mean and standard deviation (mean $\pm$ std) calculated over runs with different random seeds. The best result for each metric is highlighted in bold.

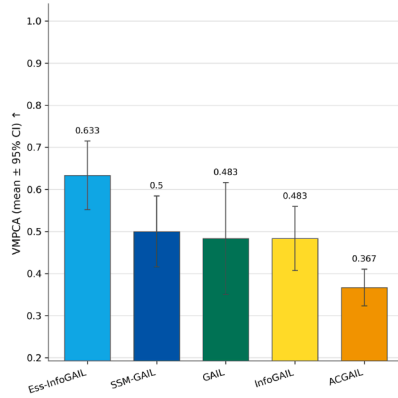
Environments	Metrics	Methods				
		GAIL	ACGAIL	InfoGAIL	Ess-InfoGAIL	SSM-GAIL
2D-Trajectory	NMI	0.469 $\pm$ 0.058	0.747 $\pm$ 0.139	0.764 $\pm$ 0.108	0.896 $\pm$ 0.045	0.933$\pm$0.043
	ENT	0.493 $\pm$ 0.037	0.309 $\pm$ 0.189	0.300 $\pm$ 0.139	0.095 $\pm$ 0.045	0.054$\pm$0.036
Reacher	NMI	0.204 $\pm$ 0.062	0.316 $\pm$ 0.027	0.336 $\pm$ 0.011	0.638 $\pm$ 0.048	0.734$\pm$0.026
	ENT	1.183 $\pm$ 0.098	1.045 $\pm$ 0.041	1.014 $\pm$ 0.031	0.638 $\pm$ 0.088	0.467$\pm$0.052
Pusher	NMI	0.685 $\pm$ 0.064	0.773 $\pm$ 0.064	0.642 $\pm$ 0.073	0.907 $\pm$ 0.073	0.942$\pm$0.040
	ENT	0.538 $\pm$ 0.127	0.467 $\pm$ 0.123	0.664 $\pm$ 0.096	0.142 $\pm$ 0.117	0.090$\pm$0.059
Walker2D	NMI	0.406 $\pm$ 0.262	0.583 $\pm$ 0.075	0.593 $\pm$ 0.078	0.769$\pm$0.116	0.690 $\pm$ 0.174
	ENT	0.346 $\pm$ 0.081	0.336 $\pm$ 0.073	0.348 $\pm$ 0.056	0.189$\pm$0.094	0.243 $\pm$ 0.115
Hopper	NMI	0.510 $\pm$ 0.274	0.733 $\pm$ 0.001	0.575 $\pm$ 0.262	0.844 $\pm$ 0.072	0.923$\pm$0.028
	ENT	0.297 $\pm$ 0.141	0.235 $\pm$ 0.069	0.197 $\pm$ 0.061	0.155 $\pm$ 0.069	0.076$\pm$0.022
PandaReach	NMI	0.718 $\pm$ 0.098	0.759 $\pm$ 0.052	0.741 $\pm$ 0.038	0.927 $\pm$ 0.080	0.955$\pm$0.027
	ENT	0.408 $\pm$ 0.110	0.395 $\pm$ 0.066	0.420 $\pm$ 0.078	0.106 $\pm$ 0.116	0.065$\pm$0.039



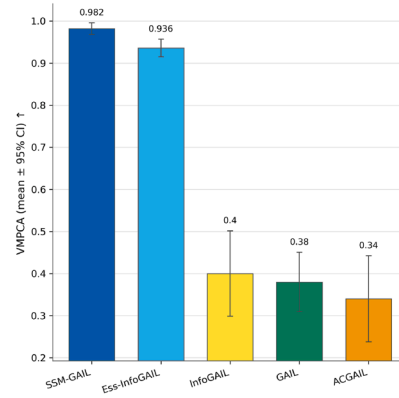
(a) Reacher



(b) Pusher



(c) Walker2D



(d) PandaReach

Fig. 4. VMPCA (mean $\pm$ 95% CI, $\uparrow$) across four environments. Bar heights denote mean values, and error bars represent 95% confidence intervals.

expert-mode prototype corresponding to the mode used to condition the rollout. Bar heights represent mean values, and error bars denote 95% confidence intervals.

In high-precision robotic arm tasks (PandaReach and Reacher), SSM-GAIL achieves higher mean VMPCA values and narrower confidence intervals, indicating stronger and more stable prototype-based video-mode consistency. On uncertain dynamics involving contact friction (Pusher), the gap between Ess-InfoGAIL and SSM-GAIL is smaller. However, on legged locomotion with strong phase transitions (Walker2D), Ess-InfoGAIL demonstrates a clear lead with lower variability, suggest-

ing that SSM-GAIL may be less effective when discrete gait phases are strongly coupled with continuous contact dynamics.

In PandaReach, SSM-GAIL achieves near-saturated VMPCA, substantially outperforming other baselines. This result suggests that temporal state-space modelling delivers stable advantages in prototype-based video-mode consistency for high-precision end-effector control. Nevertheless, Ess-InfoGAIL takes the lead in the Walker2D environment. Concurrently, all methods exhibit broad confidence intervals, indicating substantial variance across seeds for this task. Considering task characteristics, Walker2D's hybrid contact dynamics and explicit gait

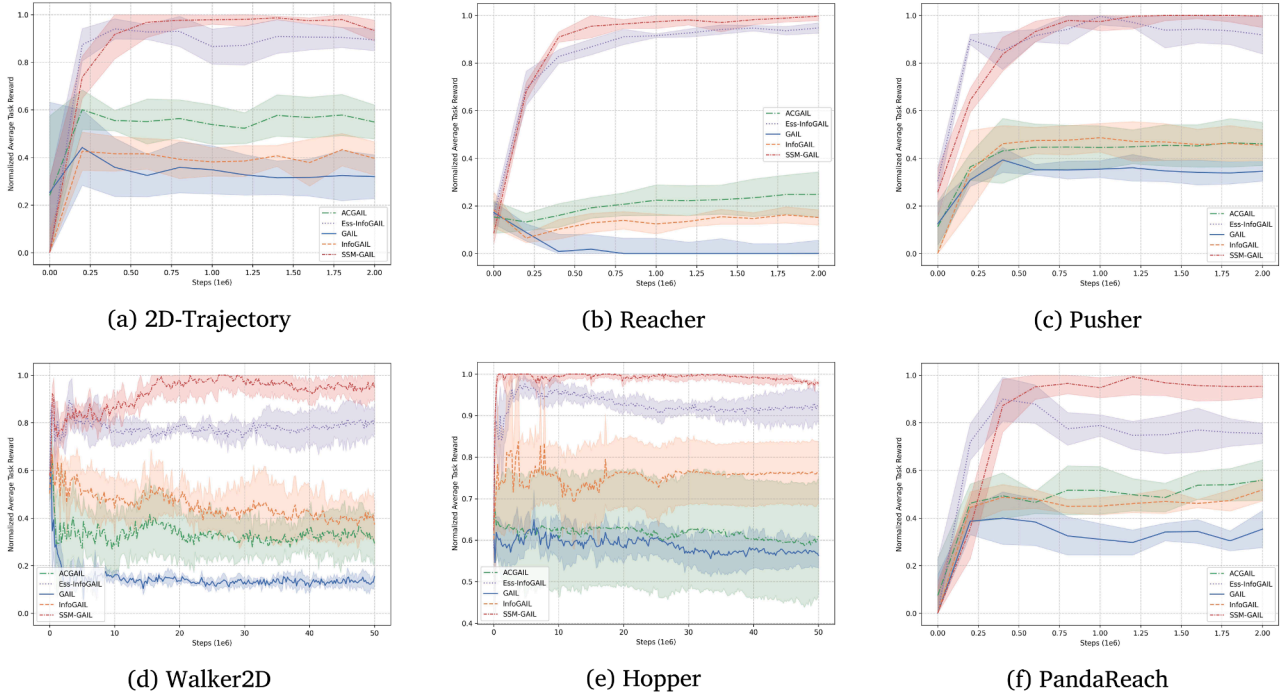


Fig. 5. Normalized average task reward curves on all six environments.

Table 3

Hyperparameter settings for training SSM-GAIL in the experiment.

Hyperparameters	Values
Surrogate loss coefficient	4.0
Discriminator gradient penalty	0.1
Value loss coefficient	5.0
Discriminator loss coefficient	20
Unsupervised loss coefficient	1.0
Unsupervised reward coefficient	0.1
Temporal reward coefficient	0.005
Policy learning rate	0.003
Value learning rate	0.003
Prior learning rate	0.003
Discriminator learning rate	0.005
Encoder learning rate	0.01
Self-supervised loss coefficient λ_{self}	0.5
Contrastive loss coefficient λ_{CT}	0.1
Contrastive temperature τ	0.1
Discount factor γ	0.99
Prediction horizon H	4

Table 4

Ablation results across six environments. The three ablation settings are: *w/o Contrastive* removes the Contrastive Learning, *w/o Pre-train* removes the Self-supervised Pre-training stage, and *w/o DSF* removes the Dynamic Scaling Factor (DSF) module. Higher NMI and lower ENT indicate better performance.

Environments	Metrics	Ablations			
		w/o Contrastive	w/o Pre-train	w/o DSF	SSM-GAIL
2D-Trajectory	NMI	0.880±0.023	0.898±0.010	0.887±0.020	0.933±0.043
	ENT	0.163±0.032	0.136±0.011	0.154±0.027	0.054±0.036
Reacher	NMI	0.699±0.045	0.714±0.073	0.709±0.039	0.734±0.026
	ENT	0.525±0.079	0.498±0.118	0.505±0.058	0.467±0.052
Pusher	NMI	0.918±0.027	0.927±0.040	0.921±0.048	0.942±0.040
	ENT	0.133±0.068	0.105±0.055	0.124±0.079	0.090±0.059
Walker2D	NMI	0.704±0.063	0.628±0.102	0.652±0.124	0.690±0.174
	ENT	0.292±0.062	0.298±0.035	0.316±0.104	0.243±0.115
Hopper	NMI	0.926±0.033	0.905±0.047	0.827±0.061	0.923±0.028
	ENT	0.079±0.035	0.095±0.044	0.157±0.050	0.076±0.022
PandaReach	NMI	0.936±0.035	0.919±0.027	0.917±0.045	0.955±0.027
	ENT	0.087±0.043	0.104±0.031	0.114±0.063	0.065±0.039

phase switching may induce phase mismatches in SSM-GAIL's smoothing prior during landing and takeoff, thereby limiting its performance. Conversely, Ess-InfoGAIL's information constraints may better separate discrete gait phases from continuous control.

Across all four environments, SSM-GAIL delivers higher prototype-based video-mode consistency and greater stability in most non-contact or weakly contact-based continuous control tasks. Conversely, in contact-dominant scenarios with pronounced gait phases, methods based on explicit information constraints may retain an advantage.

5.2.3. Mode-conditioned task reward evaluation

Reward evaluation provides a complementary view of policy performance beyond representation-level mode separation. NMI and ENT evaluate whether the learned latent representations preserve distinct behavior modes, whereas VMPCA provides a post-hoc video-level assessment of whether generated rollout videos are assigned to the expert-mode prototypes corresponding to their conditioning modes.

Task reward instead measures whether the policy can realize the corresponding control objective when conditioned on a given mode. Therefore, the reward curves are interpreted as mode-conditioned task-performance results rather than as a standalone measure of per-mode imitation fidelity.

Following the evaluation protocol used in multi-modal adversarial imitation learning (Fu et al., 2023), we evaluate policies in a modified environment with mode-specific evaluation rewards. During evaluation, each behavior mode k is activated by conditioning the policy on the corresponding mode-specific latent code $c^{(k)}$. The environment then computes the evaluation reward according to the task objective associated with that mode, such as reaching a mode-specific target, pushing the object to a mode-specific goal, or executing a mode-specific locomotion behavior. This evaluation reward is used only for testing and is not used for policy optimization.

Let R_k denote the evaluation reward associated with behavior mode k . Since each mode is evaluated with the same number of rollout steps,

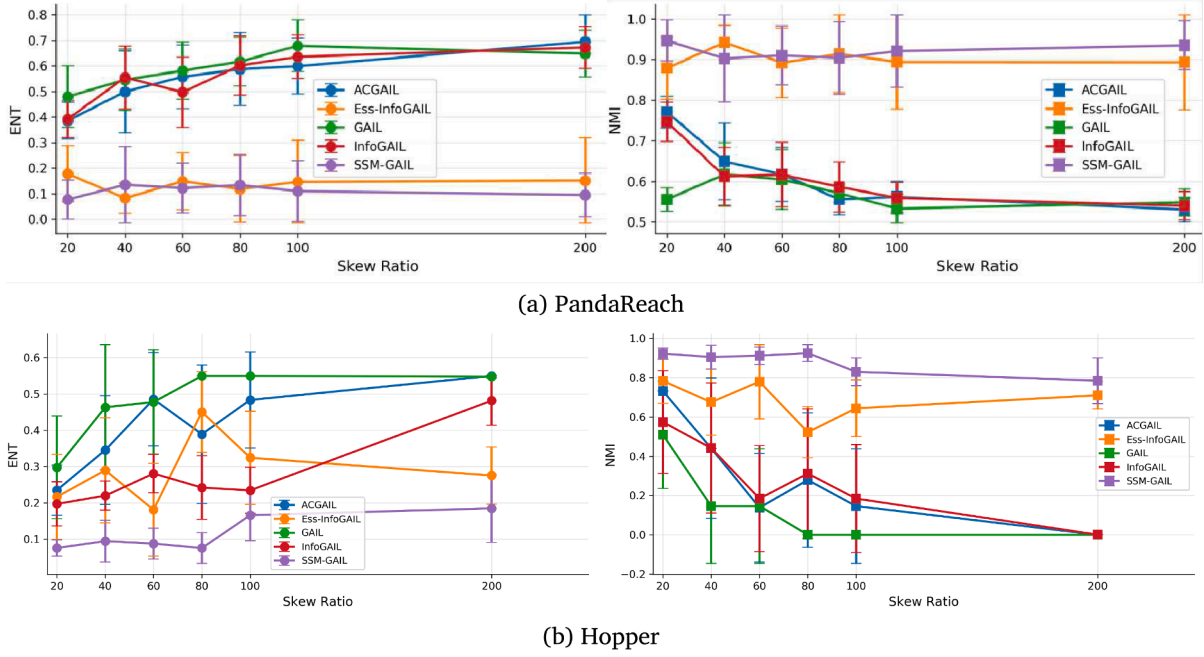


Fig. 6. Trends of ENT and NMI with respect to the skew ratio on PandaReach and Hopper. PandaReach evaluates high-precision goal-conditioned manipulation, while Hopper evaluates contact-rich locomotion with stronger nonlinear dynamics. Lower ENT and higher NMI indicate better mode purity and mode separation, respectively.

Table 5

Expert demonstration buffer statistics. Counts denote state-action transition samples. For each environment, K is the number of behavior modes. The effective skewed samples column lists the retained sample count for each mode after applying the imbalance ratio $\rho = 20$ to the large buffer.

Environment	K	Mode-identified samples per mode	Large-buffer samples per mode	Mode-identified-to-large-buffer ratio	Effective skewed samples per mode	Total effective skewed samples
2D-Trajectory	4	200	20,000	1:100	20,000 / 20,000 / 4472 / 999	45,471
Reacher	6	500	100,000	1:200	100,000 / 100,000 / 22,360 / 22,360 / 4999 / 4999	254,718
Pusher	6	500	100,000	1:200	100,000 / 100,000 / 22,360 / 22,360 / 4999 / 4999	254,718
Walker2D	3	500	100,000	1:200	100,000 / 22,360 / 4,999	127,359
Hopper	3	500	100,000	1:200	100,000 / 22,360 / 4,999	127,359
PandaReach	5	500	100,000	1:200	100,000 / 100,000 / 22,360 / 22,360 / 4999	249,719

Table 6

Network architecture parameters used in SSM-GAIL.

Module	Architecture	Activation / Output
Policy network π_θ	MLP: $(d_s + K + 1) \rightarrow 64 \rightarrow 64 \rightarrow d_a$	Tanh hidden activation; action output dimension d_a
Value network	MLP: $(d_s + K + 1) \rightarrow 64 \rightarrow 64 \rightarrow 1$	Tanh hidden activation; scalar value output
Discriminator feature extractor h_ϕ	MLP: $(d_s + d_a) \rightarrow 100 \rightarrow 100$	Tanh hidden activation; feature dimension 100
Discriminator logit head	Linear: $100 \rightarrow 1$	Discriminator logit
Mode posterior head Q	Linear: $100 \rightarrow K$	Softmax over K behavior modes
Latent-noise encoder	Linear: $100 \rightarrow 1$	Continuous latent-noise prediction
Dynamics projection	Linear: $100 \rightarrow 100$	Projected dynamic feature
Self-supervised decoder (parameterized by ψ_{dec})	MLP: $100 \rightarrow 128 \rightarrow H d_s$	ReLU hidden activation; output reshaped to $H \times d_s$

the mode-conditioned task reward is computed as

$$J_k(\pi_\theta) = \frac{T_{\max}}{N_{\text{rep}} T_{\text{eval}}} \sum_{n=1}^{N_{\text{rep}}} \sum_{t=0}^{T_{\text{eval}}-1} R_k(s_t^{(n)}, a_t^{(n)}), \quad (32)$$

where N_{rep} is the number of repeated evaluation runs, T_{eval} is the number of evaluation steps for each mode, and $T_{\max}$ is the maximum episode length used to rescale the average per-step reward to an episode-level value. The mode-averaged task reward is then obtained by averaging

over all behavior modes:

$$J_{\text{avg}}(\pi_\theta) = \frac{1}{K} \sum_{k=1}^K J_k(\pi_\theta). \quad (33)$$

The normalized average task reward curves computed from this mode-averaged evaluation reward are shown in Fig. 5(a)–(f). The compared methods include ACGAIL, Ess-InfoGAIL, GAIL, InfoGAIL, and our proposed SSM-GAIL. As shown in the figures, SSM-GAIL consistently demonstrates the fastest learning speed and achieves the highest or

Table 7

Sensitivity analysis of the prediction horizon H for SSM-GAIL. Results are summarized over six benchmark environments. For each environment, results are averaged over three random seeds; the table reports the mean and standard deviation across environment-level averages.

H	NMI	ENT
1	0.796±0.088	0.263±0.112
2	0.729±0.127	0.327±0.143
4	0.863±0.118	0.166±0.163
6	0.719±0.155	0.329±0.172
8	0.793±0.115	0.271±0.190

near-highest normalized rewards throughout training. Compared to the baselines, SSM-GAIL maintains superior performance with lower variance across random seeds, as indicated by the narrow shaded regions in the plots. While ACGAIL and Ess-InfoGAIL generally perform better than GAIL and InfoGAIL, they still fail to match the overall reward levels and stability of SSM-GAIL. In contrast, GAIL and InfoGAIL exhibit slower learning, lower final rewards, and greater instability, particularly in more challenging tasks. The baselines may perform worse due to limited ability to model diverse modes and temporal dependencies. These results indicate that SSM-GAIL achieves strong mode-conditioned task performance across diverse continuous-control environments. Together with NMI, ENT, and VMPCA, the reward evaluation provides complementary evidence that the learned policies not only separate behavior modes but also realize the corresponding task objectives.

5.2.4. Skew of the trajectory distribution

To investigate the effect of trajectory distribution skew on multi-modal imitation learning, we conduct skew-ratio experiments in two representative continuous-control environments: PandaReach and Hopper. PandaReach represents high-precision goal-conditioned manipulation, where different behavior modes correspond to different target-reaching directions. Hopper represents contact-rich locomotion with stronger nonlinear dynamics and instability, providing a more challenging setting for evaluating robustness under skewed demonstrations. For both environments, we use six skew-ratio settings and evaluate representation-level mode separation using ENT and NMI.

As shown in Fig. 6, SSM-GAIL maintains strong robustness under increasing trajectory distribution skew in both environments. In PandaReach, SSM-GAIL generally achieves low ENT and high NMI across most skew ratios, indicating stable mode purity and mode separation in high-precision manipulation. Although Ess-InfoGAIL reaches a high NMI at some moderate skew ratios, its ENT remains higher than that of SSM-GAIL, suggesting weaker mode purity.

In Hopper, the advantage of SSM-GAIL becomes more evident under contact-rich dynamics. As the skew ratio increases, several baselines show clear degradation in NMI and increased ENT, indicating difficulty in preserving underrepresented behavior modes. By contrast, SSM-GAIL maintains consistently low ENT and high NMI across most skew-ratio settings. These results show that the robustness of SSM-GAIL to skewed demonstrations is not limited to high-precision pose-control tasks, but also extends to more complex locomotion settings with nonlinear contact dynamics.

The improved robustness can be attributed to the complementary effects of self-supervised dynamics representation and frequency-aware DSF reweighting. The self-supervised objective encourages the discriminator representation to encode dynamics-relevant temporal information, while DSF increases the contribution of underrepresented modes when the demonstration distribution becomes more imbalanced. Together, these components help SSM-GAIL preserve both mode separation and mode purity under severe trajectory distribution skew.

5.2.5. Ablation analysis

Table 4 presents the experimental results for six benchmark environments after removing different modules, comparing the impact of each module on the model. Across all experiments involving the three ablation groups, NMI generally increases as ENT decreases. This indicates that as code assignability improves, code assignment entropy concurrently decreases, thereby validating the consistency between the two metrics. The table reveals that without DSF, NMI decreased by an average of 0.044 while ENT increased by 0.063; without pre-training, NMI decreased by 0.031 and ENT increased by 0.040; without contrastive learning, NMI decreased by 0.019 and ENT increased by 0.047.

Removing contrastive learning has the most pronounced effect on low-contact/low-nonlinearity tasks (2D-Trajectory, Reacher, Pusher). The degradation upon removing DSF is particularly significant on Hopper, with comprehensive deterioration also observed across other strongly dynamic/contact-based or high-dimensional pose tasks such as Walker2D and PandaReach. This indicates that DSF is important for reweighting the Q -based mode-consistency reward by mode frequency. It helps preserve underrepresented modes under strong dynamics and high-dimensional pose variations. For Walker2D, the ablation results reveal a more task-dependent behavior of different modules under contact-rich locomotion dynamics. Removing self-supervised pre-training or DSF lowers NMI and increases ENT, suggesting that future-state prediction and frequency-aware reweighting still contribute to representation stability in contact-rich locomotion. In contrast, removing contrastive learning slightly improves NMI but increases ENT. This indicates that contrastive regularization helps reduce mode-assignment uncertainty, but may also impose overly tight feature clustering when discrete gait phases and continuous contact transitions are strongly coupled.

These results suggest that the limitation on Walker2D is related to the mismatch between smooth future-state reconstruction and hybrid contact-phase dynamics, rather than the failure of a single module. A possible extension is to make the self-supervised prediction target contact-aware, for example by jointly predicting future states with contact-phase indicators, foot-ground contact states, or force-related auxiliary variables. Such contact-aware auxiliary targets may better align the learned representation with high-complexity locomotion dynamics.

5.3. Sensitivity to prediction horizon

We further analyze the effect of the prediction horizon H in the self-supervised pre-training stage. The horizon determines the temporal extent of the future-state reconstruction target and therefore affects the type of dynamics encoded in the discriminator representation. We vary H in $\{1, 2, 4, 6, 8\}$ while keeping all other hyperparameters unchanged. The resulting representation quality is evaluated using NMI and ENT, which measure behavior-mode separation and mode purity, respectively.

Table 7 reports the sensitivity results averaged over six benchmark environments. For each environment and each horizon, results are first averaged over three random seeds, and the table reports the mean and standard deviation across the resulting environment-level averages. When H is too small, the auxiliary prediction task mainly captures local transition patterns and provides limited temporal context for distinguishing behavior modes. Increasing H to a moderate value improves the temporal informativeness of the learned representation. However, further increasing H enlarges the reconstruction target and makes the auxiliary task harder to optimize, which can weaken representation quality.

Overall, $H = 4$ achieves the highest average NMI and the lowest average ENT among all tested horizons, indicating the best overall trade-off between temporal informativeness and reconstruction difficulty. Per-environment results further show that $H = 4$ performs best in most environments, including 2D-Trajectory, Hopper, Pusher, and

Table 8

Per-environment sensitivity analysis of the prediction horizon H for SSM-GAIL. Results are reported as mean $\pm$ standard deviation over three random seeds.

Environment	Metric	$H = 1$	$H = 2$	$H = 4$	$H = 6$	$H = 8$
2D-Trajectory	NMI	0.853 $\pm$ 0.047	0.770 $\pm$ 0.039	0.933$\pm$0.043	0.823 $\pm$ 0.029	0.782 $\pm$ 0.048
	ENT	0.196 $\pm$ 0.062	0.301 $\pm$ 0.059	0.054$\pm$0.036	0.226 $\pm$ 0.020	0.279 $\pm$ 0.057
Reacher	NMI	0.741$\pm$0.032	0.674 $\pm$ 0.021	0.734 $\pm$ 0.026	0.668 $\pm$ 0.057	0.635 $\pm$ 0.040
	ENT	0.447$\pm$0.055	0.588 $\pm$ 0.034	0.467 $\pm$ 0.052	0.581 $\pm$ 0.091	0.630 $\pm$ 0.070
Pusher	NMI	0.855 $\pm$ 0.035	0.882 $\pm$ 0.068	0.942$\pm$0.040	0.802 $\pm$ 0.062	0.895 $\pm$ 0.027
	ENT	0.261 $\pm$ 0.018	0.198 $\pm$ 0.089	0.090$\pm$0.059	0.261 $\pm$ 0.103	0.189 $\pm$ 0.054
Walker2D	NMI	0.836$\pm$0.165	0.571 $\pm$ 0.238	0.690 $\pm$ 0.174	0.596 $\pm$ 0.043	0.734 $\pm$ 0.151
	ENT	0.151$\pm$0.134	0.328 $\pm$ 0.125	0.243 $\pm$ 0.115	0.310 $\pm$ 0.029	0.220 $\pm$ 0.078
Hopper	NMI	0.640 $\pm$ 0.058	0.618 $\pm$ 0.068	0.923$\pm$0.028	0.506 $\pm$ 0.089	0.758 $\pm$ 0.117
	ENT	0.336 $\pm$ 0.033	0.350 $\pm$ 0.012	0.076$\pm$0.022	0.479 $\pm$ 0.082	0.240 $\pm$ 0.096
PandaReach	NMI	0.848 $\pm$ 0.178	0.856 $\pm$ 0.192	0.955$\pm$0.027	0.919 $\pm$ 0.135	0.953 $\pm$ 0.048
	ENT	0.184 $\pm$ 0.220	0.196 $\pm$ 0.279	0.065$\pm$0.039	0.116 $\pm$ 0.192	0.066 $\pm$ 0.063

PandaReach, while shorter horizons can be more favorable in some contact-rich or phase-switching locomotion tasks such as Walker2D. This suggests that the optimal horizon can be task-dependent, but $H = 4$ provides a stable default setting across diverse multi-modal control tasks. Detailed per-environment sensitivity results are provided in [Appendix B](#).

6. Conclusion

In this work, we proposed SSM-GAIL, a novel imitation learning framework that leverages self-supervised pre-training and contrastive learning to robustly disentangle and imitate diverse behavior modes from highly skewed expert demonstrations. Extensive experiments in benchmark environments demonstrate that SSM-GAIL generally outperforms baseline methods across most tasks and remains competitive under severe distribution skewness.

To complement the state-based evaluation, we additionally conduct a post-hoc assessment using Video-Mode Prototype Consistency Accuracy (VMPCA), computed by assigning generated videos to their nearest expert-mode prototypes in a pretrained R3D-18 video feature space, without modifying the training procedure. The VMPCA results provide complementary evidence that the generated rollout videos are frequently assigned to expert-mode prototypes corresponding to their conditioning modes.

Our results highlight the importance of addressing distribution imbalance for practical imitation learning, paving the way for more reliable and generalizable policy learning in real-world settings. While SSM-GAIL demonstrates strong performance, it currently relies on a small amount of mode-identified data to stabilize mode-aware representation learning. In future work, we plan to explore automatic mode discovery and further enhance the sample efficiency and scalability of our framework for more challenging and complex domains.

CRedit authorship contribution statement

Yawen Zhao: Writing – original draft, Validation, Methodology, Investigation, Data curation, Formal analysis; **Yue Chen:** Visualization, Data curation; **Fei Zhu:** Writing – review & editing, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Reproducibility details

A.1. Expert demonstration buffers

[Table 5](#) provides the detailed demonstration-buffer statistics used in our experiments. In our implementation, demonstrations are serialized at the state-action transition-sample level rather than as separate episode files. Therefore, we report the exact number of stored transition samples for each mode. Expert demonstrations are first collected separately for each behavior mode and stored as mode-specific buffers. The small mode-identified buffer provides mode-supervised signals, while the larger buffer is used to construct the skewed imitation-learning set.

For the skewed training buffer, the imbalance ratio is set to $\rho = 20$ in the main experiments. Given a stored buffer size N per mode, modes are divided into three frequency groups with effective sample counts approximately proportional to N , $N/\sqrt{\rho}$, and N/ρ , respectively. The exact effective counts follow the integer truncation used in the implementation.

A.2. Network architectures

[Table 6](#) summarizes the network architectures used in SSM-GAIL. The input and output dimensions depend on the environment-specific observation and action dimensions, while the hidden-layer widths are fixed across environments. Let d_s denote the raw observation dimension, d_a denote the action dimension, K denote the number of behavior modes, and H denote the self-supervised prediction horizon. In the main experiments, $H = 4$.

Appendix B. Additional sensitivity results

[Table 8](#) reports the per-environment sensitivity results for the prediction horizon H . The main text reports the average results across all environments, while this table provides the detailed results for each benchmark task.

References

- Allahyani, M., Alsulami, R., Alwafi, T., Alafif, T., Ammar, H., Sabban, S., & Chen, X. (2023). DivGAN: A diversity enforcing generative adversarial network for mode collapse reduction. *Artificial Intelligence*, 317, 103863.
- Beaudry, N. J., & Renner, R. (2012). An intuitive proof of the data processing inequality. *Quantum Information and Computation*, 12(5-6), 432–441.
- Boyapati, M., & Aygun, R. (2025). BalancerGNN: Balancer graph neural networks for imbalanced datasets: A case study on fraud detection. *Neural Networks*, 182, 106926.
- Cai, R., Jiang, Z., Zheng, K., Li, Z., Chen, W., Chen, X., Shen, Y., Chen, G., Hao, Z., & Zhang, K. (2025). Learning disentangled representation for multi-modal time-series sensing signals. In *Proceedings of the ACM on web conference 2025* (pp. 3247–3266).
- Darrin, M., Formont, P., Cheung, J. C. K., & Piantanida, P. (2024). Cosmic: Mutual information for task-agnostic summarization evaluation. In *Proceedings of the 62nd annual meeting of the association for computational linguistics (volume 1: Long papers)* (pp. 12696–12717).

- Fei, C., Wang, B., Zhuang, Y., Zhang, Z., Hao, J., Zhang, H., Ji, X., & Liu, W. (2020). Triple-GAIL: A multi-modal imitation learning framework with generative adversarial nets. In *Proceedings of the twenty-ninth international joint conference on artificial intelligence, IJCAI-20* (pp. 2929–2935).
- Foster, D. J., Block, A., & Misra, D. (2024). Is behavior cloning all you need? understanding horizon in imitation learning. *Advances in Neural Information Processing Systems*, 37, 120602–120666.
- Fu, H., Tang, K., Lu, Y., Qi, Y., Deng, G., Sung, F., & Chen, C. (2023). Ess-infoGAIL: Semi-supervised imitation learning from imbalanced demonstrations. *Advances in Neural Information Processing Systems*, 36, 60048–60059.
- Gallouédec, Q., Cazin, N., Dellandréa, E., & Chen, L. (2021). Panda-gym: Open-source goal-conditioned environments for robotic learning. In *4th robot learning workshop: Self-supervised and lifelong learning at neurIPS*.
- Goodfellow, I. J., Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, A., & Bengio, Y. (2014). Generative adversarial nets. *Advances in Neural Information Processing Systems*, 27, 2672–2680.
- Gu, S., & Zhu, F. (2024). Bagail: Multi-modal imitation learning from imbalanced demonstrations. *Neural Networks*, 174, 106251.
- Han, T. S., & Verdú, S. (1994). Generalizing the Fano inequality. *IEEE Transactions on Information Theory*, 40(4), 1247–1251.
- Ho, J., & Ermon, S. (2016). Generative adversarial imitation learning. *Advances in Neural Information Processing Systems*, 29, 4565–4573.
- Huang, D., Xiong, X., Ma, J., Li, J., Jie, Z., Ma, L., & Li, G. (2024). Alignsam: Aligning segment anything model to open context via reinforcement learning. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 3205–3215).
- Jiang, H., & Nachum, O. (2020). Identifying and correcting label bias in machine learning. In *International conference on artificial intelligence and statistics* (pp. 702–712). PMLR.
- Koromilas, P., Bouritsas, G., Giannakopoulos, T., Nicolaou, M., & Panagakis, Y. (2024). Bridging mini-batch and asymptotic analysis in contrastive learning: From infoNCE to kernel-based losses. In *International conference on machine learning* (pp. 25276–25301). PMLR.
- Kuefler, A., & Kochenderfer, M. J. (2018). Burn-in demonstrations for multi-modal imitation learning. In *Proceedings of the 17th international conference on autonomous agents and multiagent systems* (pp. 1071–1078).
- Li, Y., Song, J., & Ermon, S. (2017). Infogail: Interpretable imitation learning from visual demonstrations. *Advances in Neural Information Processing Systems*, 30.
- Lin, J., & Zhang, Z. (2018). Acgail: Imitation learning about multiple intentions with auxiliary classifier gans. In *Pacific rim international conference on artificial intelligence* (pp. 321–334). Springer.
- Liu, M., Lin, F., Zhang, Z., Jia, Y., & Cui, J. (2024). An enhanced driving trajectory prediction method based on generative adversarial imitation learning. In *Advanced Intelligent Computing Technology and Applications* (pp. 179–190). Springer.
- Lu, Y., Liu, A., Yu, D., Li, J., & King, I. (2024). An entropy-based text watermarking detection method. In *Proceedings of the 62nd annual meeting of the association for computational linguistics (volume 1: Long papers)* (pp. 11724–11735).
- Luo, T., Pearce, T., Chen, H., Chen, J., & Zhu, J. (2024). C-GAIL: Stabilizing generative adversarial imitation learning with control theory. *Advances in Neural Information Processing Systems*, 37, 29464–29488.
- Mildenberger, D., Hager, P., Rueckert, D., & Menten, M. J. (2025). A tale of two classes: Adapting supervised contrastive learning to binary imbalanced datasets. In *Proceedings of the computer vision and pattern recognition conference* (pp. 10305–10314).
- Ng, A. Y., & Russell, S. J. (2000). Algorithms for inverse reinforcement learning. In *Proceedings of the seventeenth international conference on machine learning* (pp. 663–670).
- Park, S., & Levine, S. (2023). Predictable mdp abstraction for unsupervised model-based rl. In *International conference on machine learning* (pp. 27246–27268). PMLR.
- Poole, B., Ozair, S., Van Den Oord, A., Alemi, A., & Tucker, G. (2019). On variational bounds of mutual information. In *International conference on machine learning* (pp. 5171–5180). PMLR.
- Seker, M. Y., Ahmetoglu, A., Nagai, Y., Asada, M., Oztup, E., & Ugur, E. (2022). Imitation and mirror systems in robots through deep modality blending networks. *Neural Networks*, 146, 22–35.
- Tang, C., Abbatematteo, B., Hu, J., Chandra, R., Martín-Martín, R., & Stone, P. (2025). Deep reinforcement learning for robotics: A survey of real-world successes. In *Proceedings of the AAAI conference on artificial intelligence* (pp. 28694–28698).
- Thabtah, F., Hammoud, S., Kamalov, F., & Gonsalves, A. (2020). Data imbalance in classification: Experimental evaluation. *Information Sciences*, 513, 429–441.
- Todorov, E., Erez, T., & Tassa, Y. (2012). Mujoco: A physics engine for model-based control. In *2012 IEEE/RSJ International conference on intelligent robots and systems* (pp. 5026–5033). IEEE.
- Wang, P., Xu, D., Fan, Z., Wang, D., Mohan, S., Iandola, F., Ranjan, R., Li, Y., Liu, Q., Wang, Z. et al. (2024). Taming mode collapse in score distillation for text-to-3d generation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 9037–9047).
- Wang, Z., Merel, J. S., Reed, S. E., Wayne, G., de Freitas, N., & Heess, N. (2017). Robust imitation of diverse behaviors. In *Advances in Neural Information Processing Systems*, 30, 5320–5329.
- Xu, T., Zhang, Z., Chen, R., Sun, Y., & Yu, Y. (2024). Provably and practically efficient adversarial imitation learning with general function approximation. *Advances in Neural Information Processing Systems*, 37, 66108–66146.
- Xu, Y., Zhang, T., & Fang, Y. (2026). Aegis: A domain generalization framework for medical image segmentation by mitigating feature misalignment. *Pattern Recognition*, 172, 112406.
- Yang, K., Yu, Z., Chen, W., Liang, Z., & Chen, C. L. P. (2024). Solving the imbalanced problem by metric learning and oversampling. *IEEE Transactions on Knowledge and Data Engineering*, 36(12), 9294–9307.
- Yang, Y., & Xu, Z. (2020). Rethinking the value of labels for improving class-imbalanced learning. *Advances in Neural Information Processing Systems*, 33, 19290–19301.
- Yao, H., Song, Z., Zhou, Y., Ao, T., Chen, B., & Liu, L. (2024). Moconvq: Unified physics-based motion control via scalable discrete representations. *ACM Transactions on Graphics (TOG)*, 43(4), 1–21.
- Zhao, Z., Yang, C., & Wu, Q. (2023). Twowin-SOVA: Two windows discrete cosine transform and synthetic minority oversampling technique one-versus-all ensemble classifiers for imbalanced hyperspectral image explainable classification. *IEEE Transactions on Geoscience and Remote Sensing*, 61, 1–16.